

Question number	Answer	Mark
3(a)	An answer that includes any two of the following points: • risk of growing {bacteria / fungi} (1) • {growth regulators / plant tissue} may cause allergic reaction (1) • sharp instruments / other sensible risk (1)	(2)

Question number	Answer	Mark
3(b)	A description that includes any three of the following points: • find suitable range of concentration of growth regulator (1) • find suitable method for measuring amylase activity (1) • find the time taken for amylase production (1) • identify {other / named} variable that needs to be taken into account (1)	(3)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes ten of the following points: • appropriate measurement of dependent variable (1) • measure the dependent variable several times and calculate a mean (1) • at least five concentrations of growth regulator (1) • description of how growth regulator is applied (1) • description of using the endosperm (1) • reference to aseptic conditions (1) • stated time period for incubation (1) • description of using starch as a substrate (1) • description of using iodine solution (1) • repeats at each concentration and mean calculated (1) • control of one variable relating to the cereal grains (1) • control of one other standardised variable (1)	For example, measuring diameter of clear zone Accept description of aseptic methods	(10)

Question number	Answer	Additional guidance	Mark
3(d)	A description that includes the following points: • table with headings (1) • means calculated from repeats (1) • {scatter / line} graph format with labelled axes (1) • use of an appropriate statistical test (1)	For example, (Pearson's) correlation coefficient or Spearman's rank	(4)

Question number	Answer	Mark
3(e)	An answer that includes any three of the following points: • difficult to control {all variables / or a named variable} (1) • another factor may be limiting effect of growth regulator (1) • possible contamination with {bacteria / fungi} (1) • more than one growth regulator may be involved (1)	(3)